Thermal Properties of Matter

Comprehensive Theory, Thermometry, Thermal Expansion, Calorimetry, Phase Transitions, Heat Transfer, and Exam-Oriented Practice

1. Learning Objectives & Chapter Overview

In everyday experience, we use the terms *hot* and *cold* as relative indicators of the thermal state of an object. A kettle of boiling water is hotter than a box containing ice. In physics, however, sensory perception by touch is too limited and unreliable for quantitative work, requiring rigorous definitions of **temperature** and **heat**. This tutorial develops the complete framework of thermal physics required for the **NEET UG** examination:

- Distinguish clearly between temperature (a measure of hotness) and heat (energy transferred solely due to a temperature difference).
- Calibrate and convert between Celsius (°C), Fahrenheit (°F), Kelvin (K), and custom thermometric scales using fixed reference points and the triple point of water.
- Apply the ideal gas laws to constant-volume gas thermometry and understand the physical origin of absolute zero ($-273.15\ ^\circ\mathrm{C}$ or $0\ \mathrm{K}$).
- Analyze linear (α_l), area ($\alpha_A = 2\alpha_l$), and volume ($\alpha_v = 3\alpha_l$) thermal expansion of solids, liquids, and gases, including the anomalous expansion of water between $0\ ^\circ\mathrm{C}$ and $4\ ^\circ\mathrm{C}$ and thermal stress in rigidly fixed rods.
- Master specific heat capacity, molar heat capacities (C_p and C_v), calorimetry, phase changes, regelation, and latent heats of fusion (L_f) and vaporisation (L_v).
- Solve quantitative problems involving steady-state heat conduction across single and composite slabs, natural and forced convection, blackbody radiation (Wien's displacement law and Stefan-Boltzmann law), and Newton's law of cooling.

2. Temperature, Heat, and Thermometry

2.1 Temperature vs. Heat

Temperature is a relative measure or quantitative indication of the degree of hotness or coldness of a body. When a glass of ice-cold water is left on a table on a hot summer day, it eventually warms up, whereas a cup of hot tea on the same table cools down. Whenever a temperature difference exists between a system and its surrounding medium, energy flows until both attain the same temperature (thermal equilibrium).

Heat is defined as the form of energy transferred between two (or more) systems, or between a system and its surroundings, *by virtue of a temperature difference*. For ice-cold water, heat flows from the environment into the glass; for hot tea, heat flows from the cup to the environment.

- **SI Unit of Heat Energy (ΔQ):** joule (J); dimensional formula $[\mathrm{ML^2T^{-2}}]$. Another common non-SI unit is the calorie (cal), where $1\ \mathrm{cal} \approx 4.186\ \mathrm{J}$. Note that heat Q is a path-dependent energy transfer, *not* a state variable.
- **SI Unit of Temperature (T):** kelvin (K); dimensional formula $[\mathrm{K}]$. A commonly used practical unit is degree Celsius ($^\circ\mathrm{C}$).

2.2 Measurement of Temperature & Temperature Scales

A thermometer utilizes a measurable physical property of matter—called a **thermometric property**—that varies sufficiently and predictably with temperature. Common liquid-in-glass thermometers use mercury or alcohol, whose volume varies linearly with temperature over a wide range. Electrical resistance thermometers use the linear variation of resistance R with temperature.

To construct any standard temperature scale, two fixed reference points correlated to reproducible physical phenomena are chosen. Historically, the **ice point** (freezing point of pure water under standard atmospheric pressure) and the **steam point** (boiling point of pure water under standard atmospheric pressure) were selected:

- **Celsius Scale (t_C):** Ice point $= 0\ ^\circ\mathrm{C}$, Steam point $= 100\ ^\circ\mathrm{C}$, divided into 100 equal intervals.
- **Fahrenheit Scale (t_F):** Ice point $= 32\ ^\circ\mathrm{F}$, Steam point $= 212\ ^\circ\mathrm{F}$, divided into 180 equal intervals.

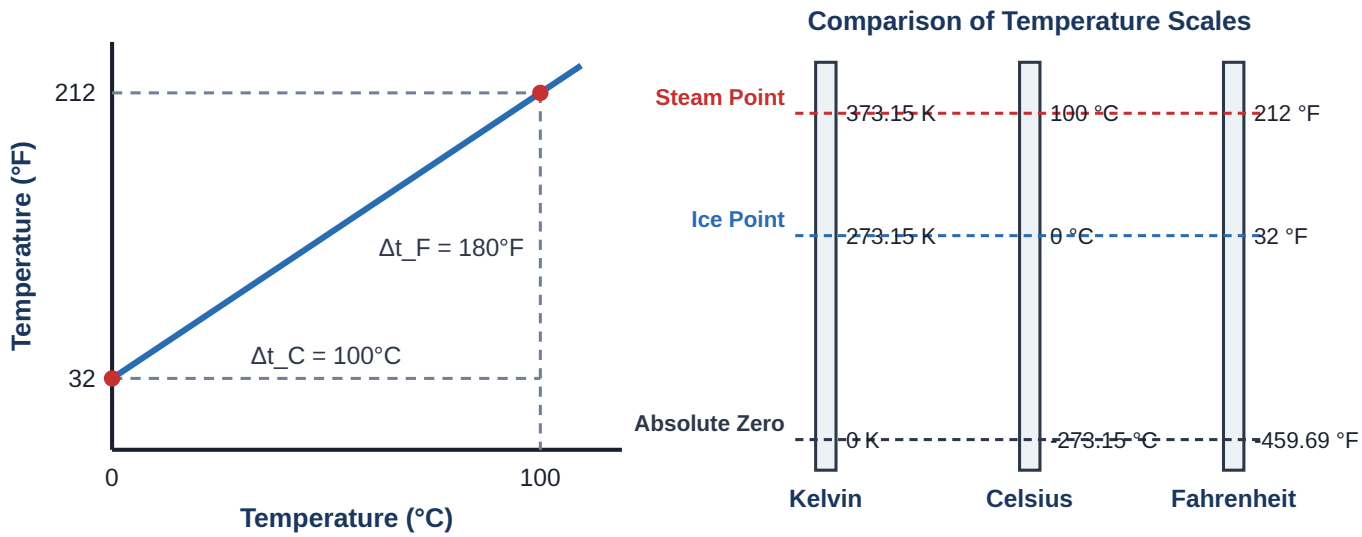
Celsius–Fahrenheit Conversion Relation:

$$\frac{t_F - 32}{180} = \frac{t_C}{100} \implies t_F = \frac{9}{5}\,t_C + 32$$

For a small or finite change in temperature:

$$\Delta t_F = \frac{9}{5}\,\Delta t_C = 1.8\,\Delta t_C$$

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Tutorial Diagram 1: Left — Linear relationship between Fahrenheit and Celsius scales. Right — Alignment of Steam Point, Ice Point, and Absolute Zero across Kelvin, Celsius, and Fahrenheit scales.

2.3 Ideal-Gas Equation, Constant-Volume Gas Thermometer & Absolute Temperature

Liquid-in-glass thermometers constructed with different liquids (e.g., mercury vs. alcohol) agree at the fixed points but show slightly different readings at intermediate temperatures because expansion properties differ from liquid to liquid. Experiments show, however, that **all gases at low densities exhibit the exact same expansion behaviour**:

- **Boyle's Law (constant T):** $PV = \text{constant}$
- **Charles' Law (constant P):** $\dfrac{V}{T} = \text{constant}$

Combining these two laws for μ moles of any low-density gas yields the **Ideal-Gas Equation**:

$$\frac{PV}{T} = \mu R \implies PV = \mu RT$$

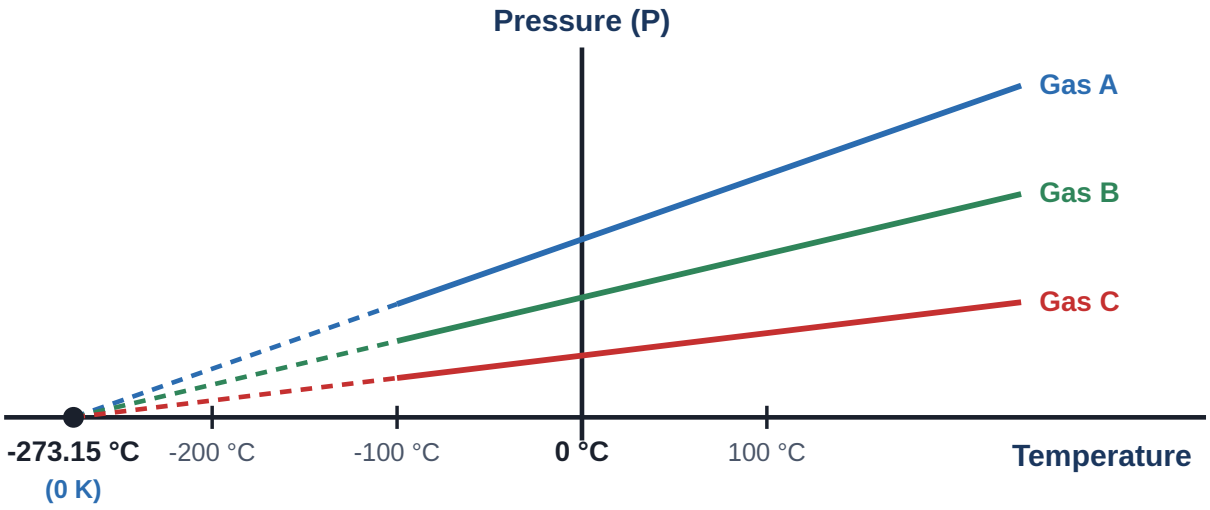
where μ is the number of moles in the gas sample and $R = 8.31\,\mathrm{J\,mol^{-1}\,K^{-1}}$ is the universal gas constant.

In a **constant-volume gas thermometer**, the volume V of a low-density gas is held constant, so $P \propto T$. Temperature is read directly in terms of pressure. When pressure P is plotted against temperature t_C for different low-density gases, straight lines are obtained over a wide temperature range. Extrapolating these straight lines to the zero-pressure axis reveals that all low-density gases converge to the exact same minimum temperature: $-273.15\,^{\circ}\mathrm{C}$.

This extrapolated zero-pressure temperature is designated as **absolute zero** $(0\,\mathrm{K})$—the foundation of the **Kelvin (absolute) temperature scale**, where molecular activity reaches its theoretical minimum. Choosing the unit interval size on the Kelvin scale to be identical to $1\,^{\circ}\mathrm{C}$ gives the exact conversion:

$$T = t_C + 273.15 \quad \text{or} \quad t_C = T - 273.15$$

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Tutorial Diagram 2: Pressure versus temperature for three different low-density gases held at constant volume. Extrapolation to zero pressure defines absolute zero (-273.15 °C or 0 K).

Modern Thermometry & The Triple Point of Water:
Why is the triple point of water preferred over the melting point of ice and boiling point of water as the standard fixed point in modern thermometry? Because the melting and boiling points depend significantly on external atmospheric pressure. In contrast, the **triple point of water**—the unique state at which solid ice, liquid water, and water vapour coexist in thermal equilibrium—occurs at a single, unique pressure ($6.11 \times 10^{-3}\,\mathrm{atm}$ or $611\,\mathrm{Pa}$, also noted in the chapter box as $0.006\,\mathrm{atm}$) and a unique temperature assigned the exact value:

$$T_{\mathrm{tr}} = 273.16\,\mathrm{K} \quad (\text{which corresponds to } t_C = 0.01\,^{\circ}\mathrm{C})$$

The two fixed points of the modern Kelvin scale are therefore **absolute zero** ($0\,\mathrm{K}$) and the **triple point of water** ($273.16\,\mathrm{K}$). Since the ice point at $1\,\mathrm{atm}$ is $0.01\,\mathrm{K}$ below the triple point of water, $0\,^{\circ}\mathrm{C}$ corresponds to $273.16 - 0.01 = 273.15\,\mathrm{K}$, which explains why 273.15 (and not 273.16) appears in $t_C = T - 273.15$.For any constant-volume gas thermometer calibrated at the triple point:

$$T = \frac{P}{P_{\mathrm{tr}}} \times 273.16\,\mathrm{K}$$

3. Thermal Expansion of Solids, Liquids, and Gases

Most substances expand when heated and contract when cooled. A tightly screwed metallic lid on a sealed glass bottle can be loosened by placing the lid in hot water because the metal expands more than the glass. The increase in dimensions of a body due to an increase in its temperature is called **thermal expansion**.

3.1 Linear, Area, and Volume Expansion

- **Linear Expansion:** For a long rod of initial length l subjected to a small temperature change ΔT, the fractional change in length $\dfrac{\Delta l}{l}$ is directly proportional to ΔT:

$$\frac{\Delta l}{l} = \alpha_l\,\Delta T \implies l' = l(1 + \alpha_l\,\Delta T)$$

  where α_l is the **coefficient of linear expansion** (or linear expansivity), with SI unit K^{-1} or $^{\circ}\mathrm{C}^{-1}$.
- **Area (Superficial) Expansion:** For a solid sheet of area A, the fractional change in area for a small temperature change ΔT is:

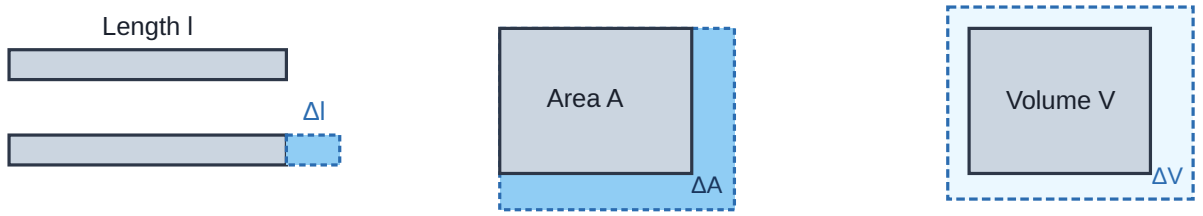
$$\frac{\Delta A}{A} = \alpha_A\,\Delta T = 2\alpha_l\,\Delta T$$

- **Volume (Cubical) Expansion:** For a 3D body or fluid of volume V, the fractional change in volume for a temperature change ΔT defines the **coefficient of volume expansion** α_v:

$$\alpha_v = \left(\frac{\Delta V}{V}\right)\frac{1}{\Delta T} \implies \frac{\Delta V}{V} = \alpha_v\,\Delta T = 3\alpha_l\,\Delta T$$

  Note: α_v is not strictly constant at all temperatures; for a metal like copper, α_v increases with temperature at low temperatures and becomes constant only at high temperatures (above roughly $500\,\mathrm{K}$).

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(a) Linear Expansion

Δl / l = α_l ΔT

(b) Area Expansion

ΔA / A = 2α_l ΔT

(c) Volume Expansion

ΔV / V = 3α_l ΔT

Tutorial Diagram 3: Geometric comparison of linear, area, and volume thermal expansion of isotropic solids.

Proof of $\alpha_v = 3\alpha_l$ for an Isotropic Solid

Consider a solid cube of side length l and initial volume $V = l^3$ that expands equally in all three mutually perpendicular directions when its temperature increases by ΔT. Since $\Delta l = \alpha_l l \Delta T$, the new volume is:

$$V + \Delta V = (l + \Delta l)^3 = l^3 + 3l^2\Delta l + 3l(\Delta l)^2 + (\Delta l)^3$$

Because $\Delta l \ll l$ for typical temperature changes (since $\alpha_l \sim 10^{-5}\,\mathrm{K}^{-1}$), terms containing $(\Delta l)^2$ and $(\Delta l)^3$ are negligible and may be dropped:

$$\Delta V \simeq 3l^2\Delta l = 3l^2(\alpha_l l \Delta T) = 3V\alpha_l \Delta T \implies \alpha_v = \left(\frac{\Delta V}{V}\right)\frac{1}{\Delta T} = 3\alpha_l$$

Thus, the ratio of the three expansion coefficients for an isotropic solid is $\alpha_l : \alpha_A : \alpha_v = 1 : 2 : 3$. Note also that when a solid containing a cavity or a drilled hole is heated, the dimensions of the hole or cavity expand at the exact same rate as the solid material itself.

Reference Data: Coefficients of Linear and Volume Expansion ($0\,^\circ\mathrm{C}$ to $100\,^\circ\mathrm{C}$)

Material (Solids)	α_l ($10^{-5}\,\mathrm{K}^{-1}$)	Substance (Solids & Liquids)	α_v (K^{-1})
Aluminium	2.5	Aluminium	7×10^{-5}
Brass	1.8	Brass	6×10^{-5}
Iron	1.2	Iron	3.55×10^{-5}
Copper	1.7	Paraffin	58.8×10^{-5}
Silver	1.9	Glass (ordinary)	2.5×10^{-5}
Gold	1.4	Glass (pyrex)	1×10^{-5}
Glass (pyrex)	0.32	Hard rubber / Invar	2.4×10^{-4} / 2×10^{-6}
Lead	2.9	Mercury / Water / Ethanol	18.2×10^{-5} / 20.7×10^{-5} / 110×10^{-5}

Key Observations: Copper expands about five times more than Pyrex glass for the same temperature rise. Invar (an iron-nickel alloy) and Pyrex glass have exceptionally low expansivities. Ethanol expands much more than mercury for the same rise in temperature.

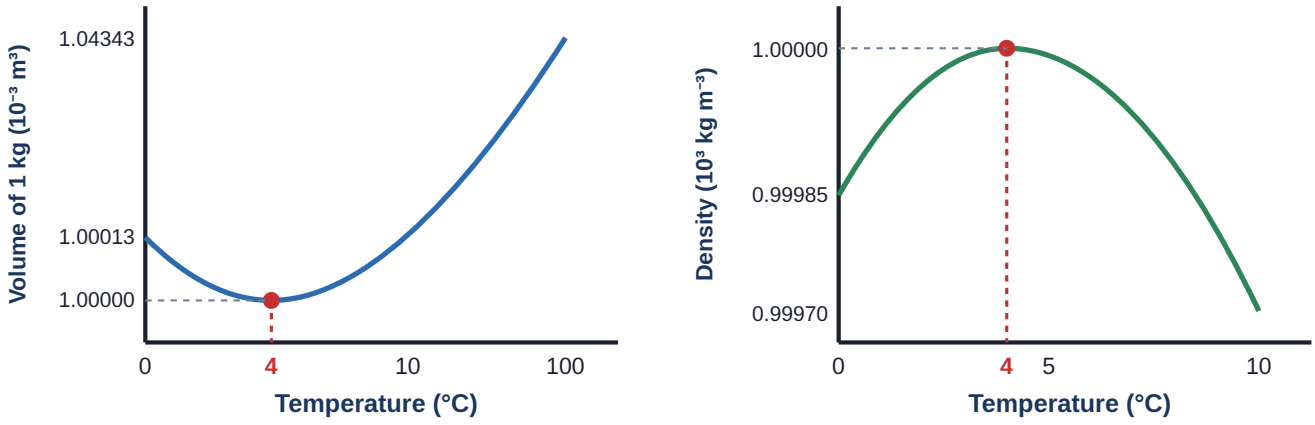
3.2 Variation of Density with Temperature & Anomalous Expansion of Water

For a fixed mass m of a substance, density is $\rho = \frac{m}{V}$. Since mass remains constant during heating, $\frac{\Delta\rho}{\rho} = -\frac{\Delta V}{V} = -\alpha_v\,\Delta T$, or $\rho' \simeq \rho(1 - \alpha_v\,\Delta T)$.

Anomalous Behaviour of Water: Unlike normal liquids, water **contracts on heating between $0\,^\circ\mathrm{C}$ and $4\,^\circ\mathrm{C}$.** When water is cooled from room temperature, its volume decreases until it reaches $4\,^\circ\mathrm{C}$. Below $4\,^\circ\mathrm{C}$ (from $4\,^\circ\mathrm{C}$ down to $0\,^\circ\mathrm{C}$), its volume increases and therefore its density decreases. Consequently, **water has its minimum volume and maximum density ($1.00000 \times 10^3\,\mathrm{kg\,m^{-3}}$) at $4\,^\circ\mathrm{C}$.**

Environmental Significance: Because of this anomalous property, lakes and ponds in cold climates freeze at the top surface first. As a lake cools toward $4\,^\circ\mathrm{C}$, surface water loses heat to the atmosphere, becomes denser, and sinks, while warmer water rises. Once the water cools below $4\,^\circ\mathrm{C}$, however, colder water becomes *less dense* and stays at the top surface, where it eventually freezes into a protective layer of ice while aquatic animal and plant life survive in the $4\,^\circ\mathrm{C}$ liquid water below.

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Tutorial Diagram 4: Anomalous expansion of water. Left — Volume of 1 kg of water reaches a minimum at 4 °C. Right — Density of water reaches a maximum at 4 °C.

3.3 Volume Expansion of an Ideal Gas & Thermal Stress

Ideal Gas Expansion at Constant Pressure: From $PV = \mu RT$, at constant pressure $P\,\Delta V = \mu R\,\Delta T$. Dividing the two relations gives:

$$\frac{\Delta V}{V} = \frac{\Delta T}{T} \implies \alpha_v = \left(\frac{\Delta V}{V}\right)\frac{1}{\Delta T} = \frac{1}{T} \quad \text{(for an ideal gas at constant pressure)}$$

At $0\,^\circ\mathrm{C}$ ($T = 273.15\,\mathrm{K}$), $\alpha_v = 3.7 \times 10^{-3}\,\mathrm{K}^{-1}$, and at room temperature ($\sim 300\,\mathrm{K}$), $\alpha_v \approx 3300 \times 10^{-6}\,\mathrm{K}^{-1}$, which is orders of magnitude larger than for solids and liquids and decreases with increasing temperature.

Thermal Stress in a Rigidly Clamped Rod: If a rod of length l, cross-sectional area A, Young's modulus Y, and linear expansivity α_l is rigidly clamped at both ends so that expansion or contraction is prevented when temperature changes by ΔT, a compressive or tensile strain $\dfrac{\Delta l}{l} = \alpha_l\,\Delta T$ is developed. The resulting **thermal stress** and thermal force ΔF are:

$$\text{Thermal Stress} = \frac{\Delta F}{A} = Y\left(\frac{\Delta l}{l}\right) = Y\,\alpha_l\,\Delta T, \qquad \Delta F = YA\alpha_l\,\Delta T$$

For example, a steel rail of length $5\,\mathrm{m}$ and cross-section $A = 40\,\mathrm{cm}^2 = 40 \times 10^{-4}\,\mathrm{m}^2$ ($\alpha_l = 1.2 \times 10^{-5}\,\mathrm{K}^{-1}$, $Y = 2 \times 10^{11}\,\mathrm{N\,m}^{-2}$) heated by $\Delta T = 10\,^\circ\mathrm{C}$ develops a strain of 1.2×10^{-4}, a thermal stress of $2.4 \times 10^{7}\,\mathrm{N\,m}^{-2}$, and a force $\Delta F \simeq 10^5\,\mathrm{N}$ capable of bending the rails if no expansion gaps are provided.

4. Specific Heat Capacity, Calorimetry, and Change of State

4.1 Heat Capacity, Specific Heat Capacity, and Molar Specific Heat

Experiments show that the quantity of heat ΔQ required to raise the temperature of a body by ΔT (without phase change) is proportional to its mass m and to ΔT, and depends on the nature of the substance:

- **Heat Capacity (S):** $S = \dfrac{\Delta Q}{\Delta T}$ [SI Unit: $\mathrm{J\,K}^{-1}$]
- **Specific Heat Capacity (s):** $s = \dfrac{S}{m} = \dfrac{1}{m}\dfrac{\Delta Q}{\Delta T} \implies \Delta Q = ms\Delta T$ [SI Unit: $\mathrm{J\,kg}^{-1}\,\mathrm{K}^{-1}$, Dimensions: $[\mathrm{L}^2\mathrm{T}^{-2}\mathrm{K}^{-1}]$]
- **Molar Specific Heat Capacity (C):** For μ moles, $C = \dfrac{S}{\mu} = \dfrac{1}{\mu}\dfrac{\Delta Q}{\Delta T} \implies \Delta Q = \mu C\Delta T$ [SI Unit: $\mathrm{J\,mol}^{-1}\,\mathrm{K}^{-1}$]

For gases, heat transfer conditions must be specified: C_p is the molar specific heat capacity at constant pressure, and C_v is the molar specific heat capacity at constant volume.

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Substance	Specific Heat s ($\mathrm{J\,kg^{-1}\,K^{-1}}$)	Gas	C_p ($\mathrm{J\,mol^{-1}\,K^{-1}}$)	C_v ($\mathrm{J\,mol^{-1}\,K^{-1}}$)
Aluminium / Carbon	900.0 / 506.5	Helium (He)	20.8	12.5
Copper / Iron	386.4 / 450	Hydrogen ($\mathrm{H_2}$)	28.8	20.4
Lead / Silver / Tungsten	127.7 / 236.1 / 134.4	Nitrogen ($\mathrm{N_2}$)	29.1	20.8
Ice / Glass / Mercury	2060 / 840 / 140	Oxygen ($\mathrm{O_2}$)	29.4	21.1
Kerosene / Edible oil	2118 / 1965	Carbon dioxide ($\mathrm{CO_2}$)	37.0	28.5
Water	4186.0	*Water has the highest specific heat capacity among common substances.*		

Applications of Water's High Specific Heat Capacity: Water is used as a coolant in automobile radiators and as a heating medium in hot water bags. Because water warms up and cools down much more slowly than land, coastal regions experience moderate climates while desert areas heat up rapidly by day and cool rapidly at night.

4.2 Calorimetry

Calorimetry is the measurement of heat. A **calorimeter** consists of a metallic vessel and stirrer (usually copper or aluminium) placed inside a wooden jacket packed with heat-insulating material like glass wool, with a thermometer inserted through an opening in the lid. For an **isolated system** (no heat exchange with the surroundings), when bodies at different temperatures are mixed:

> **Principle of Calorimetry (Conservation of Heat Energy):**
> $$\text{Heat Lost by Hotter Part(s)} = \text{Heat Gained by Colder Part(s)}$$

4.3 Change of State (Phase Transitions), Triple Point & Latent Heat

During a phase transition (change of state), **temperature remains constant** even though heat is continuously supplied or removed, and both phases coexist in thermal equilibrium:

- **Melting (Fusion) & Freezing:** Solid $\leftrightarrow$ Liquid transition at the **melting point**. The melting point at $1\,\mathrm{atm}$ is the *normal melting point*.
- **Regelation:** Melting of ice under increased pressure at a temperature below $0\,^\circ\mathrm{C}$, followed by refreezing once the pressure is removed. If a metallic wire with heavy weights attached to its ends is placed across an ice slab, the high pressure beneath the wire lowers the melting point of ice, causing the ice directly below the wire to melt; as the wire passes through, the water above it refreezes, so the wire cuts completely through the ice slab without splitting it into two pieces. Skating on snow is possible because pressure under the skates forms a thin lubricating film of water.
- **Vaporisation & Boiling:** Liquid $\leftrightarrow$ Vapour (gas) transition at the **boiling point**. Increasing pressure raises the boiling point (e.g., inside a pressure cooker, making cooking faster). Decreasing pressure lowers the boiling point (e.g., at high hill altitudes where atmospheric pressure is low, making cooking difficult).
- **Sublimation:** Direct transition between solid and vapour states without passing through the liquid state (e.g., dry ice [solid $\mathrm{CO_2}$] and iodine).

Phase Diagrams (P–T Diagrams) and Triple Point

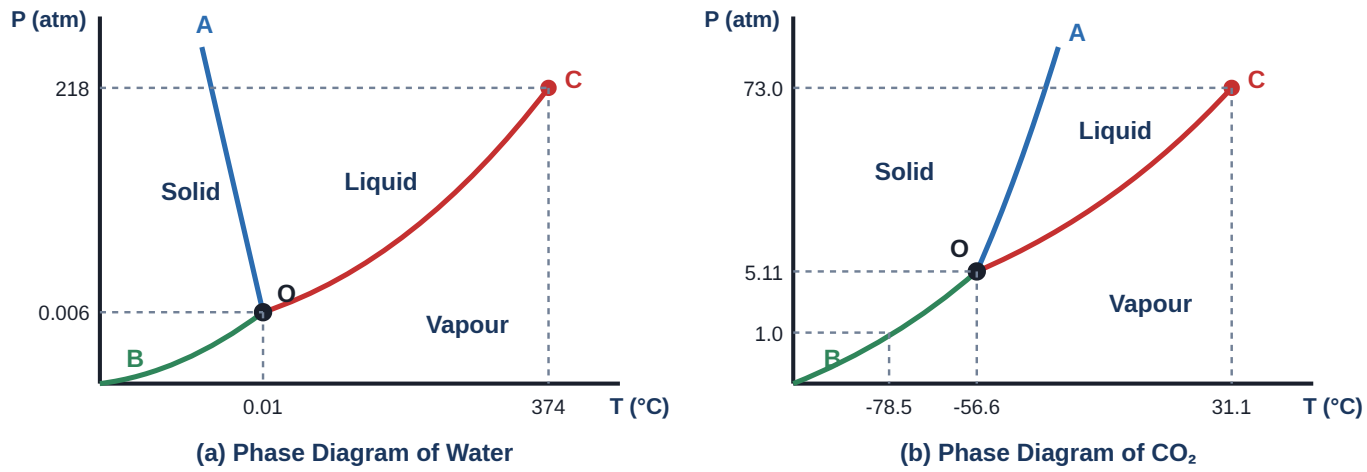
A pressure–temperature (P–T) phase diagram divides the plane into solid, liquid, and vapour regions separated by three coexistence curves:

1. **Fusion Curve (AO):** Solid and liquid coexist in equilibrium. (For water, the fusion curve slopes leftward, showing that melting point decreases with pressure; for $\mathrm{CO_2}$, it slopes rightward.)
2. **Sublimation Curve (BO):** Solid and vapour coexist in equilibrium.
3. **Vaporisation Curve (CO):** Liquid and vapour coexist in equilibrium (terminating at the critical point C).

The unique point O where all three curves meet and all three phases (solid, liquid, vapour) coexist in equilibrium is the **triple point**:

- **Water ($\mathrm{H_2O}$):** Triple point at $T = 273.16\,\mathrm{K}$ ($0.01\,^\circ\mathrm{C}$) and $P = 0.006\,\mathrm{atm}$ ($6.11 \times 10^{-3}\,\mathrm{atm}$ or $611\,\mathrm{Pa}$). Critical point at $374\,^\circ\mathrm{C}$ and $218\,\mathrm{atm}$.
- **Carbon Dioxide ($\mathrm{CO_2}$):** Triple point at $T = -56.6\,^\circ\mathrm{C}$ ($216.55\,\mathrm{K}$) and $P = 5.11\,\mathrm{atm}$. Because liquid $\mathrm{CO_2}$ cannot exist at $1.0\,\mathrm{atm}$ (which lies below $5.11\,\mathrm{atm}$), solid $\mathrm{CO_2}$ sublimes directly to vapour at $-78.5\,^\circ\mathrm{C}$ at $1.0\,\mathrm{atm}$.

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Tutorial Diagram 5: Pressure–Temperature phase diagrams (not to scale) for (a) Water and (b) Carbon dioxide, showing the Fusion Curve (AO), Sublimation Curve (BO), Vaporisation Curve (CO), and Triple Point (O).

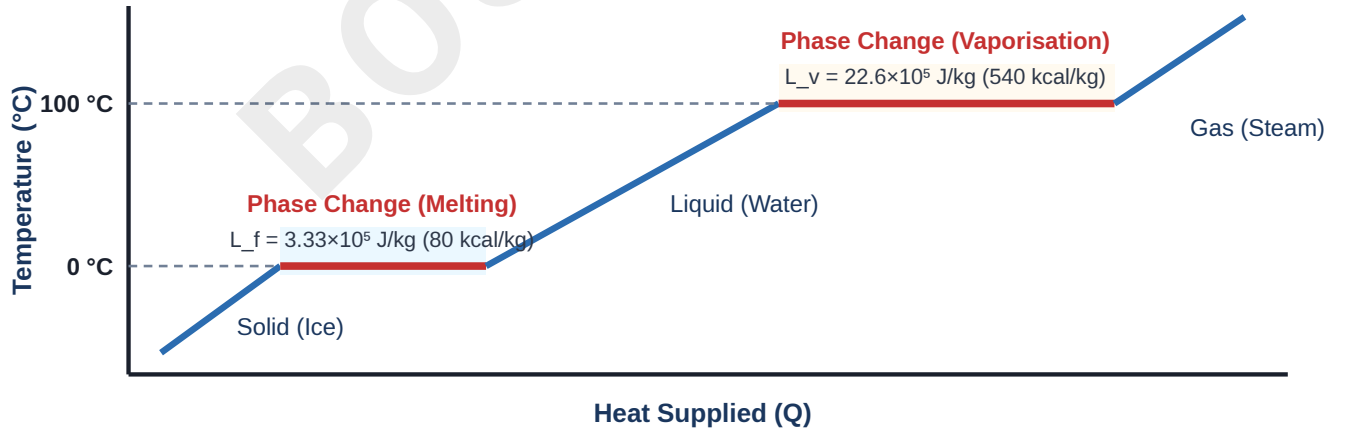
Latent Heat (L)

The amount of heat energy transferred per unit mass during a change of state of a substance at constant temperature and pressure is called its **latent heat** L:

$$Q = mL \implies L = \frac{Q}{m} \qquad [\text{SI Unit: J kg}^{-1},\ \text{Dimensions: } [\text{L}^2\text{T}^{-2}]]$$

- **Latent heat of fusion of water (L_f):** $L_f = 3.33 \times 10^5\ \text{J kg}^{-1}$ (or $3.35 \times 10^5\ \text{J kg}^{-1} \approx 80\ \text{kcal kg}^{-1}$) at $0\ ^\circ\text{C}$ and $1\ \text{atm}$.
- **Latent heat of vaporisation of water (L_v):** $L_v = 22.6 \times 10^5\ \text{J kg}^{-1}$ (or $2.256 \times 10^6\ \text{J kg}^{-1} \approx 540\ \text{kcal kg}^{-1}$) at $100\ ^\circ\text{C}$ and $1\ \text{atm}$.

Why steam burns are more severe than boiling water burns: Steam at $100\ ^\circ\text{C}$ carries $22.6 \times 10^5\ \text{J kg}^{-1}$ more heat energy than liquid water at $100\ ^\circ\text{C}$ due to its large latent heat of vaporisation.



Tutorial Diagram 6: Temperature versus heat supplied for water at 1 atm pressure (not to scale). Horizontal plateaus represent constant-temperature phase transitions; different slopes of the inclined lines reflect different specific heat capacities of ice, water, and steam.

5. Modes of Heat Transfer & Newton's Law of Cooling

Heat energy is transferred from regions of higher temperature to regions of lower temperature by three distinct mechanisms: **conduction**, **convection**, and **radiation**.

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5.1 Conduction

Conduction is the transfer of heat between adjacent parts of a body through molecular collisions without any bulk flow of matter. Solids (especially metals) are the best thermal conductors, gases are poor conductors, and liquids have intermediate conductivities.

Consider a uniform bar of length L and cross-sectional area A whose sides are thermally insulated and whose ends are maintained at steady temperatures T_C and T_D ($T_C > T_D$). In the **steady state**, the temperature decreases uniformly along the bar, and the time rate of heat flow (**heat current** H) through any cross-section is:

$$H = \frac{\Delta Q}{\Delta t} = KA\frac{T_C - T_D}{L} = -KA\frac{\mathrm{d}T}{\mathrm{d}x}$$

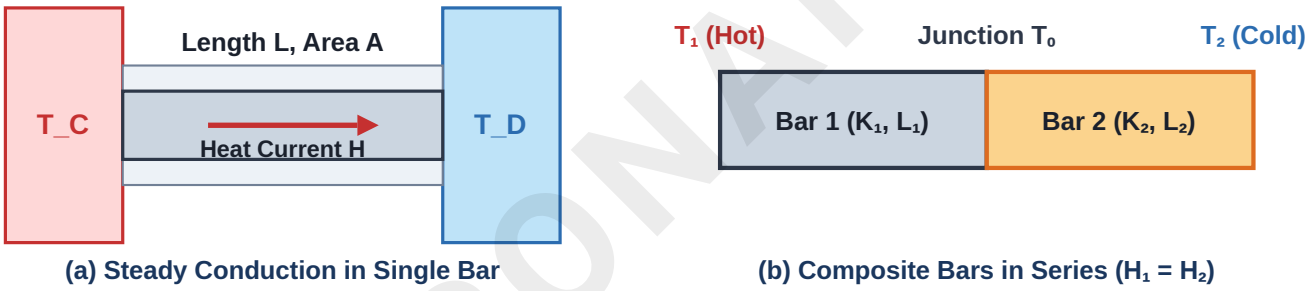
where K is the **thermal conductivity** of the material [SI unit: $\mathrm{J\,s^{-1}\,m^{-1}\,K^{-1}}$ or $\mathrm{W\,m^{-1}\,K^{-1}}$; dimensions: $[\mathrm{MLT^{-3}K^{-1}}]$].

Thermal Resistance & Composite Bars in Series: By analogy with Ohm's law ($I = \Delta V/R$), we can define thermal resistance $R_{\mathrm{th}} = \dfrac{L}{KA}$ so that $H = \dfrac{\Delta T}{R_{\mathrm{th}}}$. When two insulated bars of lengths L_1, L_2, areas A_1, A_2, and conductivities K_1, K_2 are joined end-to-end in series between temperatures T_1 and T_2, the steady-state heat current H is identical through both bars:

$$H = \frac{K_1A_1(T_1 - T_0)}{L_1} = \frac{K_2A_2(T_0 - T_2)}{L_2}$$

If $A_1 = A_2 = A$ and $L_1 = L_2 = L$, the junction temperature T_0 and equivalent thermal conductivity K' of the compound bar of total length $2L$ are:

$$T_0 = \frac{K_1T_1 + K_2T_2}{K_1 + K_2}, \qquad K' = \frac{2K_1K_2}{K_1 + K_2}$$



(a) Steady Conduction in Single Bar (b) Composite Bars in Series ($H_1 = H_2$)

Tutorial Diagram 7: (a) Steady-state heat flow H across an insulated bar of length L and cross-section A. (b) Two bars connected in series with junction temperature T_0.

Metals	K ($\mathrm{W\,m^{-1}\,K^{-1}}$)	Non-metals & Gases	K ($\mathrm{W\,m^{-1}\,K^{-1}}$)
Silver	406	Ice / Concrete / Glass / Water	1.6 / 0.8 / 0.8 / 0.8
Copper	385	Body fat / Insulating brick / Wood	0.20 / 0.15 / 0.12
Aluminium / Brass	205 / 109	Felt / Glass wool	0.04 / 0.04
Steel / Lead / Mercury	50.2 / 34.7 / 8.3	Hydrogen / Air / Argon	0.14 / 0.024 / 0.016

5.2 Convection

Convection is heat transfer by the actual bulk motion of matter and is possible **only in fluids** (liquids and gases):

- **Natural Convection:** Driven by gravity and buoyancy due to density differences between heated (less dense, rising) and cooler (more dense, sinking) fluid layers. Examples include **sea breeze** during the day (land heats faster than water; warm air over land rises and cooler air from the sea blows in), **land breeze** at night (land cools faster than water, reversing the wind cycle), and global **trade winds** (modified by the Earth's rotation, where equatorial air has an eastward speed of $1600\,\mathrm{km/h}$ and descends near $30^\circ\mathrm{N}$ latitude).
- **Forced Convection:** Fluid is forced to circulate by an external pump or fan (e.g., human circulatory system where the heart pumps blood to maintain uniform body temperature, automobile engine cooling systems, forced-air home heating).

5.3 Radiation & Blackbody Radiation Laws

Radiation requires no material medium; energy is transferred as electromagnetic waves (**thermal radiation**) travelling in vacuum at the speed of light $c = 3 \times 10^8\,\mathrm{m\,s^{-1}}$. Black surfaces absorb and emit radiant energy much better than light-coloured or polished silvery surfaces. A **Dewar flask (thermos bottle)** minimizes all three modes of heat transfer: a vacuum between double

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glass walls prevents conduction and convection, silvered inner and outer walls reflect radiation, and a cork support prevents conduction.

1. Wien's Displacement Law:
Blackbody radiation has a continuous wavelength spectrum. The wavelength λ_m corresponding to maximum emitted radiation intensity is inversely proportional to absolute temperature T:

$$\lambda_m T = b = 2.9 \times 10^{-3}\,\mathrm{m\,K}$$

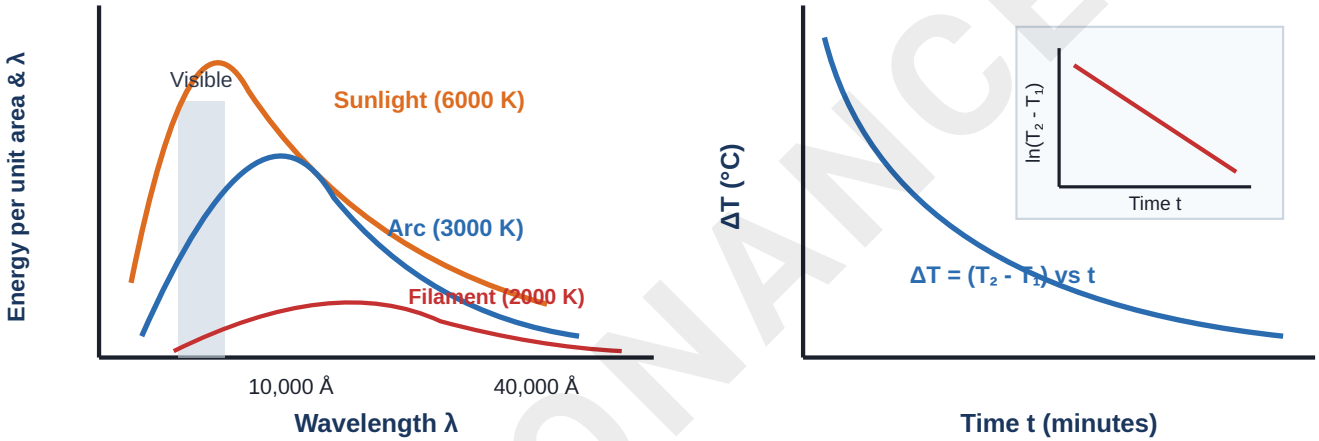
Applications: As iron is heated, λ_m shifts from infrared to dull red, reddish-yellow, and white-hot. Moon's peak radiation at $\lambda_m = 14\,\mu\mathrm{m}$ gives $T \approx 200\,\mathrm{K}$; solar radiation peaking at $\lambda_m = 4753\,\text{Å}$ gives the Sun's surface temperature $T \approx 6060\,\mathrm{K}$.

2. Stefan–Boltzmann Law:
The total electromagnetic energy radiated per unit time (H) by a body of surface area A, emissivity e ($e = 1$ for an ideal blackbody), and absolute temperature T is:

$$H_{\text{emitted}} = Ae\sigma T^4 \qquad \left(\sigma = 5.67 \times 10^{-8}\,\mathrm{W\,m^{-2}\,K^{-4}}\right)$$

When surrounded by an enclosure at absolute temperature T_s, the **net rate of radiant energy loss** is:

$$H_{\text{net}} = e\sigma A\left(T^4 - T_s^4\right)$$



Tutorial Diagram 8: Left — Blackbody radiation curves showing Wien's displacement of peak wavelength λ_m toward shorter wavelengths as temperature rises. Right — Newton's cooling curve (ΔT vs. t) and linear semi-log verification plot (ln(T₂ - T₁) vs. t).

5.4 Newton's Law of Cooling

For small temperature differences between a hot body at temperature T_2 and its surroundings at temperature T_1, the rate of loss of heat $-\dfrac{\mathrm{d}Q}{\mathrm{d}t}$ is directly proportional to the temperature excess $\Delta T = (T_2 - T_1)$:

$$-\frac{\mathrm{d}Q}{\mathrm{d}t} = k(T_2 - T_1)$$

Since $\mathrm{d}Q = ms\,\mathrm{d}T_2$, where m is the mass and s is the specific heat capacity of the cooling body:

$$-ms\frac{\mathrm{d}T_2}{\mathrm{d}t} = k(T_2 - T_1) \implies \frac{\mathrm{d}T_2}{\mathrm{d}t} = -K(T_2 - T_1) \quad \left(\text{where } K = \frac{k}{ms}\right)$$

Integrating with respect to time t gives the exact exponential decay law:

$$\log_e(T_2 - T_1) = -Kt + c \implies T_2(t) = T_1 + C'e^{-Kt}$$

Average-Temperature Approximation (Standard NEET Method):

When a body cools from T_i to T_f in time t in surroundings at T_1, we approximate $\dfrac{\mathrm{d}T_2}{\mathrm{d}t}$ by $\dfrac{T_i - T_f}{t}$ and T_2 by the arithmetic mean $T_{\text{avg}} = \dfrac{T_i + T_f}{2}$:

$$\frac{T_i - T_f}{t} = K\left(\frac{T_i + T_f}{2} - T_1\right)$$

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6. Comprehensive Worked Examples

Worked Example 1: Thermal Expansion — Fitting an Iron Ring on a Wooden Wheel

Problem: A blacksmith fixes an iron ring on the rim of the wooden wheel of a horse cart. The diameters of the wooden rim and the iron ring are $5.243\,\text{m}$ and $5.231\,\text{m}$, respectively, at $27\,^\circ\text{C}$. To what temperature should the iron ring be heated so as to fit the rim of the wheel? (Coefficient of linear expansion of iron $\alpha_l = 1.20 \times 10^{-5}\,\text{K}^{-1}$).

Solution:
Given initial temperature $T_1 = 27\,^\circ\text{C}$, initial ring diameter $L_{T_1} = 5.231\,\text{m}$, and target diameter $L_{T_2} = 5.243\,\text{m}$. Using the linear expansion relation:

$$L_{T_2} = L_{T_1}\left[1 + \alpha_l (T_2 - T_1)\right]$$

$$5.243 = 5.231\left[1 + 1.20 \times 10^{-5}(T_2 - 27)\right] \implies T_2 - 27 = \frac{5.243 - 5.231}{5.231 \times 1.20 \times 10^{-5}} = \frac{0.012}{6.2772 \times 10^{-5}} \approx 191.2\,^\circ\text{C}$$

$$T_2 = 27 + 191.2 \approx 218\,^\circ\text{C}$$

Worked Example 2: Calorimetry — Specific Heat Capacity of Aluminium

Problem: A sphere of $0.047\,\text{kg}$ aluminium is placed for sufficient time in a vessel containing boiling water so that the sphere is at $100\,^\circ\text{C}$. It is then immediately transferred to a $0.14\,\text{kg}$ copper calorimeter containing $0.25\,\text{kg}$ water at $20\,^\circ\text{C}$. The temperature of water rises and attains a steady state at $23\,^\circ\text{C}$. Calculate the specific heat capacity of aluminium. ($s_\text{w} = 4.18 \times 10^3\,\text{J kg}^{-1}\,\text{K}^{-1}$, $s_\text{Cu} = 0.386 \times 10^3\,\text{J kg}^{-1}\,\text{K}^{-1}$).

Solution:
Heat lost by aluminium sphere $= m_1 s_\text{Al} \Delta T_1 = 0.047 \times s_\text{Al} \times (100 - 23) = 3.619\, s_\text{Al}$.
Heat gained by water and copper calorimeter:

$$Q_\text{gained} = (m_2 s_\text{w} + m_3 s_\text{Cu})\Delta T_2 = (0.25 \times 4.18 \times 10^3 + 0.14 \times 0.386 \times 10^3)(23 - 20)$$

$$Q_\text{gained} = (1045 + 54.04) \times 3 = 1099.04 \times 3 = 3297.12\,\text{J}$$

Equating heat lost to heat gained:

$$3.619\, s_\text{Al} = 3297.12 \implies s_\text{Al} = \frac{3297.12}{3.619} \approx 911\,\text{J kg}^{-1}\,\text{K}^{-1} = 0.911\,\text{kJ kg}^{-1}\,\text{K}^{-1}$$

Worked Example 3: Multi-Stage Phase Transition — Ice at $-12\,^\circ\text{C}$ to Steam at $100\,^\circ\text{C}$

Problem: Calculate the total heat required to convert $3\,\text{kg}$ of ice at $-12\,^\circ\text{C}$ kept in a calorimeter to steam at $100\,^\circ\text{C}$ at atmospheric pressure. Given: $s_\text{ice} = 2100\,\text{J kg}^{-1}\,\text{K}^{-1}$, $s_\text{water} = 4186\,\text{J kg}^{-1}\,\text{K}^{-1}$, $L_{f,\text{ice}} = 3.35 \times 10^5\,\text{J kg}^{-1}$, $L_\text{steam} = 2.256 \times 10^6\,\text{J kg}^{-1}$.

Solution:
The conversion occurs in four sequential steps for $m = 3\,\text{kg}$:

1. Heat ice from $-12\,^\circ\text{C}$ to $0\,^\circ\text{C}$: $Q_1 = m s_\text{ice} \Delta T_1 = 3 \times 2100 \times 12 = 75,600\,\text{J}$.
2. Melt ice at $0\,^\circ\text{C}$ to water at $0\,^\circ\text{C}$: $Q_2 = m L_{f,\text{ice}} = 3 \times 3.35 \times 10^5 = 1,005,000\,\text{J}$.
3. Heat water from $0\,^\circ\text{C}$ to $100\,^\circ\text{C}$: $Q_3 = m s_\text{water} \Delta T_2 = 3 \times 4186 \times 100 = 1,255,800\,\text{J}$.
4. Vaporise water at $100\,^\circ\text{C}$ to steam at $100\,^\circ\text{C}$: $Q_4 = m L_\text{steam} = 3 \times 2.256 \times 10^6 = 6,768,000\,\text{J}$.

Total heat required: $Q = Q_1 + Q_2 + Q_3 + Q_4 = 9,104,400\,\text{J} \approx 9.1 \times 10^6\,\text{J}$.

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Worked Example 4: Steady-State Conduction in a Composite Rod

Problem: A steel rod of length $L_1 = 15.0\,\text{cm}$ is joined in series with a copper rod of length $L_2 = 10.0\,\text{cm}$. Their sides are thermally insulated. The free end of the steel rod is maintained in a furnace at $300\,^\circ\text{C}$ and the free end of the copper rod is in an ice box at $0\,^\circ\text{C}$. The cross-sectional area of the steel rod is twice that of the copper rod ($A_1 = 2A_2$). Find the steady-state temperature of the steel–copper junction. ($K_{\text{steel}} = 50.2\,\text{W m}^{-1}\,\text{K}^{-1}$, $K_{\text{Cu}} = 385\,\text{W m}^{-1}\,\text{K}^{-1}$).

Solution:

In the steady state, the heat current through the steel rod equals the heat current through the copper rod:

$$\frac{K_1 A_1 (300 - T)}{L_1} = \frac{K_2 A_2 (T - 0)}{L_2}$$

Substituting $A_1 = 2A_2$, $L_1 = 15.0\,\text{cm}$, $L_2 = 10.0\,\text{cm}$, $K_1 = 50.2$, and $K_2 = 385$:

$$\frac{50.2 \times 2(300 - T)}{15} = \frac{385\,T}{10} \implies 6.693(300 - T) = 38.5\,T \implies 2008 = 45.193\,T \implies T \approx 44.4\,^\circ\text{C}$$

Worked Example 5: Newton's Law of Cooling

Problem: A pan filled with hot food cools from $94\,^\circ\text{C}$ to $86\,^\circ\text{C}$ in $2\,\text{minutes}$ when the room temperature is $20\,^\circ\text{C}$. How long will it take to cool from $71\,^\circ\text{C}$ to $69\,^\circ\text{C}$?

Solution:

Case 1 ($94\,^\circ\text{C} \to 86\,^\circ\text{C}$): Average temperature $T_{\text{avg},1} = \dfrac{94 + 86}{2} = 90\,^\circ\text{C}$. Excess temperature above room temperature $\Delta T_1 = 90 - 20 = 70\,^\circ\text{C}$. Using the average form of Newton's law of cooling:

$$\frac{94 - 86}{2\,\text{min}} = K(70\,^\circ\text{C}) \implies \frac{8}{2} = 70K \quad \text{--- (1)}$$

Case 2 ($71\,^\circ\text{C} \to 69\,^\circ\text{C}$): Average temperature $T_{\text{avg},2} = \dfrac{71 + 69}{2} = 70\,^\circ\text{C}$. Excess temperature $\Delta T_2 = 70 - 20 = 50\,^\circ\text{C}$. Let the required time be t:

$$\frac{71 - 69}{t} = K(50\,^\circ\text{C}) \implies \frac{2}{t} = 50K \quad \text{--- (2)}$$

Dividing equation (1) by equation (2):

$$\frac{4}{2/t} = \frac{70}{50} \implies 2t = 1.4 \implies t = 0.7\,\text{min} = 42\,\text{s}$$

7. NEET UG Exam-Style Practice Questions (With Worked Solutions)

Note: All questions below are original practice questions designed strictly in accordance with the NEET UG syllabus and examination pattern.

BOSONANCE

Practice Question 1 — Thermometry & Custom Temperature Scales

Two absolute temperature scales A and B have the triple point of water defined to be $200\,\mathrm{A}$ and $350\,\mathrm{B}$, respectively. If the triple point of carbon dioxide is $216.55\,\mathrm{K}$, what is the relation between readings T_A and T_B on the two scales, and what is the triple point of carbon dioxide on the Fahrenheit scale?

(A) $\dfrac{T_A}{T_B} = \dfrac{4}{7}$ and $-69.88\,^\circ\mathrm{F}$

(B) $\dfrac{T_A}{T_B} = \dfrac{7}{4}$ and $-56.60\,^\circ\mathrm{F}$

(C) $\dfrac{T_A}{T_B} = \dfrac{4}{7}$ and $-101.88\,^\circ\mathrm{F}$

(D) $\dfrac{T_A}{T_B} = \dfrac{2}{3.5}$ and $-37.88\,^\circ\mathrm{F}$

Correct Answer: (A)

Worked Solution:

Since both A and B are absolute scales, their zero coincides with $0\,\mathrm{K}$. The size of $1\,\mathrm{A}$ is $\dfrac{273.16\,\mathrm{K}}{200}$ and the size of $1\,\mathrm{B}$ is $\dfrac{273.16\,\mathrm{K}}{350}$. For any given physical temperature T:

$$T = T_A\left(\frac{273.16}{200}\right) = T_B\left(\frac{273.16}{350}\right) \implies \frac{T_A}{T_B} = \frac{200}{350} = \frac{4}{7}$$

For the triple point of $\mathrm{CO_2}$ ($T = 216.55\,\mathrm{K}$):

$$t_C = T - 273.15 = 216.55 - 273.15 = -56.60\,^\circ\mathrm{C}$$

Converting to Fahrenheit:

$$t_F = \frac{9}{5}t_C + 32 = \frac{9}{5}(-56.60) + 32 = -101.88 + 32 = -69.88\,^\circ\mathrm{F}$$

Practice Question 2 — Platinum Resistance Thermometry

The electrical resistance in ohms of a certain thermometer varies with absolute temperature T according to the linear law $R = R_0[1 + \alpha(T - T_0)]$. The resistance is $101.6\,\Omega$ at the triple point of water ($273.16\,\mathrm{K}$) and $165.5\,\Omega$ at the normal melting point of lead ($600.5\,\mathrm{K}$). What is the temperature when the resistance is $123.4\,\Omega$?

(A) $345.2\,\mathrm{K}$

(B) $384.8\,\mathrm{K}$

(C) $412.6\,\mathrm{K}$

(D) $436.8\,\mathrm{K}$

Correct Answer: (B)

Worked Solution:

Taking $T_0 = 273.16\,\mathrm{K}$ and $R_0 = 101.6\,\Omega$, the linear resistance relation gives:

$$R - R_0 = R_0\alpha(T - T_0) \implies \frac{T_2 - T_0}{T_1 - T_0} = \frac{R_2 - R_0}{R_1 - R_0}$$

Given $R_1 = 165.5\,\Omega$ at $T_1 = 600.5\,\mathrm{K}$, and $R_2 = 123.4\,\Omega$ at unknown temperature T_2:

$$T_2 - 273.16 = \left(\frac{123.4 - 101.6}{165.5 - 101.6}\right)(600.5 - 273.16) = \left(\frac{21.8}{63.9}\right)(327.34) \approx 111.67\,\mathrm{K}$$

$$T_2 = 273.16 + 111.67 = 384.83\,\mathrm{K} \approx 384.8\,\mathrm{K}$$

BOSONANCE

Practice Question 3 — Constant-Volume Gas Thermometers

Two constant-volume gas thermometers A (using oxygen) and B (using hydrogen) record the following pressures:
At the triple point of water ($273.16\,\text{K}$): $P_A = 1.250 \times 10^5\,\text{Pa}$, $P_B = 0.200 \times 10^5\,\text{Pa}$.
At the normal melting point of sulphur: $P'_A = 1.797 \times 10^5\,\text{Pa}$, $P'_B = 0.287 \times 10^5\,\text{Pa}$.
What are the temperatures of the melting point of sulphur read by A and B, and how can the slight discrepancy between them be reduced?

(A) $T_A = 392.69\,\text{K}$, $T_B = 391.98\,\text{K}$; by taking measurements at still lower gas pressures/densities.

(B) $T_A = 392.69\,\text{K}$, $T_B = 391.98\,\text{K}$; by increasing the gas pressures to $10\,\text{atm}$.

(C) $T_A = 373.15\,\text{K}$, $T_B = 372.50\,\text{K}$; by replacing hydrogen with carbon dioxide.

(D) $T_A = 412.50\,\text{K}$, $T_B = 410.00\,\text{K}$; by reducing the volume of the thermometer bulbs.

Correct Answer: (A)

Worked Solution:
Using $T = \dfrac{P'}{P_{\text{tr}}} \times 273.16\,\text{K}$:

$$T_A = \frac{1.797 \times 10^5}{1.250 \times 10^5} \times 273.16 = 392.69\,\text{K}$$

$$T_B = \frac{0.287 \times 10^5}{0.200 \times 10^5} \times 273.16 = 391.98\,\text{K}$$

The slight difference occurs because oxygen and hydrogen are real gases and do not behave strictly as ideal gases at finite pressures. To reduce this discrepancy, measurements should be taken at still lower pressures (lower gas densities) where real gases approach ideal gas behaviour.

Practice Question 4 — Thermal Expansion of a Calibrated Steel Tape and Rod

A steel tape $1\,\text{m}$ long is correctly calibrated for a temperature of $27.0\,^\circ\text{C}$. The length of a steel rod measured by this tape is found to be $63.0\,\text{cm}$ on a hot day when the ambient temperature is $45.0\,^\circ\text{C}$. Given $\alpha_{\text{steel}} = 1.20 \times 10^{-5}\,\text{K}^{-1}$, what is the actual length of the steel rod at $45.0\,^\circ\text{C}$, and what is its length when the temperature returns to $27.0\,^\circ\text{C}$?

(A) $63.0136\,\text{cm}$ at $45.0\,^\circ\text{C}$; $63.0000\,\text{cm}$ at $27.0\,^\circ\text{C}$

(B) $63.0000\,\text{cm}$ at $45.0\,^\circ\text{C}$; $62.9864\,\text{cm}$ at $27.0\,^\circ\text{C}$

(C) $63.0272\,\text{cm}$ at $45.0\,^\circ\text{C}$; $63.0136\,\text{cm}$ at $27.0\,^\circ\text{C}$

(D) $62.9864\,\text{cm}$ at $45.0\,^\circ\text{C}$; $63.0000\,\text{cm}$ at $27.0\,^\circ\text{C}$

Correct Answer: (A)

Worked Solution:
At $45.0\,^\circ\text{C}$ ($\Delta T = 45.0 - 27.0 = 18.0\,^\circ\text{C}$), each $1\,\text{cm}$ mark on the steel tape has expanded to $1 \times [1 + \alpha_{\text{steel}}\Delta T]\,\text{cm}$. Hence, the actual length of the rod at $45.0\,^\circ\text{C}$ corresponding to 63.0 scale divisions is:

$$L_{45} = 63.0\left[1 + (1.20 \times 10^{-5})(18.0)\right] = 63.0(1 + 2.16 \times 10^{-4}) = 63.0136\,\text{cm}$$

Because both the rod and the tape are made of the same material (steel) and expand at the exact same rate, on a day when the temperature is $27.0\,^\circ\text{C}$ (the calibration temperature), the rod contracts by the exact same factor as the tape and its length at $27.0\,^\circ\text{C}$ is exactly $63.0\,\text{cm}$ ($63.0000\,\text{cm}$).

BOSONANCE

Practice Question 5 — Thermal Contraction of a Shaft and Thermal Expansion of a Hole

Statement I: At $27\,^\circ\text{C}$, the outer diameter of a steel shaft is $8.70\,\text{cm}$ and the diameter of the central hole in a steel wheel is $8.69\,\text{cm}$. If the shaft is cooled using dry ice ($\alpha_{\text{steel}} = 1.20 \times 10^{-5}\,\text{K}^{-1}$), the wheel slips onto the shaft when the shaft reaches approximately $-69\,^\circ\text{C}$.
Statement II: A hole of diameter $4.24\,\text{cm}$ is drilled in a copper sheet at $27.0\,^\circ\text{C}$ ($\alpha_{\text{Cu}} = 1.70 \times 10^{-5}\,\text{K}^{-1}$). When the sheet is heated to $227\,^\circ\text{C}$, the diameter of the hole decreases by $1.44 \times 10^{-2}\,\text{cm}$.
In the light of the above statements, choose the correct option:

(A) Both Statement I and Statement II are true.

(B) Both Statement I and Statement II are false.

(C) Statement I is true, but Statement II is false.

(D) Statement I is false, but Statement II is true.

Correct Answer: (C)

Worked Solution:
For Statement I: We need the shaft diameter to contract from $D_1 = 8.70\,\text{cm}$ to $D_2 = 8.69\,\text{cm}$, so $\Delta D = -0.01\,\text{cm}$:

$$\Delta T = \frac{\Delta D}{D_1 \alpha_{\text{steel}}} = \frac{-0.01}{8.70 \times 1.20 \times 10^{-5}} \approx -95.8\,^\circ\text{C} \implies T_2 = 27 - 95.8 \approx -68.8\,^\circ\text{C} \approx -69\,^\circ\text{C}$$

Thus Statement I is **true**.
For Statement II: A cavity or hole in an isotropic solid expands (never contracts) on heating at the same rate as the solid material:

$$\Delta D = D\alpha_{\text{Cu}}\Delta T = 4.24 \times (1.70 \times 10^{-5}) \times (227 - 27) = 4.24 \times 1.70 \times 10^{-5} \times 200 = +1.44 \times 10^{-2}\,\text{cm}$$

The diameter of the hole *increases* by $1.44 \times 10^{-2}\,\text{cm}$. Thus Statement II is **false**.

Practice Question 6 — Thermal Stress and Free Expansion of Joined Rods

A brass wire $1.8\,\text{m}$ long and $2.0\,\text{mm}$ in diameter at $27\,^\circ\text{C}$ is held taut with negligible initial tension between two rigid supports. If the wire is cooled to $-39\,^\circ\text{C}$, what tension is developed in the wire? Furthermore, if a brass rod of length $50\,\text{cm}$ is joined end-to-end with a steel rod of length $50\,\text{cm}$ at $40\,^\circ\text{C}$ with both outer ends *free to expand*, what is the thermal stress at the junction when heated to $250\,^\circ\text{C}$? ($\alpha_{\text{brass}} = 2.0 \times 10^{-5}\,\text{K}^{-1}$, $Y_{\text{brass}} = 0.91 \times 10^{11}\,\text{Pa}$).

(A) Tension $= 3.8 \times 10^2\,\text{N}$; Thermal stress at junction $= 0$

(B) Tension $= 3.8 \times 10^2\,\text{N}$; Thermal stress at junction $= 1.2 \times 10^8\,\text{Pa}$

(C) Tension $= 1.5 \times 10^3\,\text{N}$; Thermal stress at junction $= 0$

(D) Tension $= 7.6 \times 10^2\,\text{N}$; Thermal stress at junction $= 2.4 \times 10^7\,\text{Pa}$

Correct Answer: (A)

Worked Solution:
Part 1 (Clamped brass wire): Radius $r = 1.0\,\text{mm} = 1.0 \times 10^{-3}\,\text{m}$, area $A = \pi r^2 = 3.1416 \times 10^{-6}\,\text{m}^2$, magnitude of temperature change $\Delta T = 27 - (-39) = 66\,\text{K}$. The tensile force developed is:

$$F = YA\alpha_{\text{brass}}\Delta T = (0.91 \times 10^{11}) \times (3.1416 \times 10^{-6}) \times (2.0 \times 10^{-5}) \times 66 \approx 377.4\,\text{N} \approx 3.8 \times 10^2\,\text{N}$$

Part 2 (Free ends): Because the outer ends of the combined brass–steel rod are completely free to expand, no external constraining force acts on the rods, so **no thermal stress** is developed at the junction (total elongation is simply $\Delta l_{\text{brass}} + \Delta l_{\text{steel}} = 0.21\,\text{cm} + 0.126\,\text{cm} = 0.34\,\text{cm}$).

BOSONANCE

Practice Question 7 — Fractional Change in Density & Mechanical Power Heating

The coefficient of volume expansion of glycerine is $49 \times 10^{-5}\,\mathrm{K}^{-1}$. What is the magnitude of the fractional decrease in its density for a $30\,^{\circ}\mathrm{C}$ rise in temperature? In a separate experiment, a $10\,\mathrm{kW}$ drilling machine drills a bore in an $8.0\,\mathrm{kg}$ aluminium block ($s_{\mathrm{Al}} = 0.91\,\mathrm{J\,g^{-1}\,K^{-1}}$) for $2.5\,\mathrm{minutes}$. Assuming 50% of the mechanical power goes into heating the block, what is the rise in temperature of the block?

(A) 0.0147 and $103\,^{\circ}\mathrm{C}$

(B) 0.0147 and $206\,^{\circ}\mathrm{C}$

(C) 0.0294 and $103\,^{\circ}\mathrm{C}$

(D) 0.0049 and $51.5\,^{\circ}\mathrm{C}$

Correct Answer: (A)

Worked Solution:

1. Fractional decrease in density of glycerine:

$$\left|\frac{\Delta\rho}{\rho_0}\right| \simeq \alpha_v \Delta T = (49 \times 10^{-5}\,\mathrm{K}^{-1}) \times 30\,\mathrm{K} = 1.47 \times 10^{-2} = 0.0147$$

2. Temperature rise of aluminium block:

Total energy delivered by the $10\,\mathrm{kW}$ drill in $t = 2.5\,\mathrm{min} = 150\,\mathrm{s}$ is $W = Pt = 10 \times 10^3 \times 150 = 1.5 \times 10^6\,\mathrm{J}$. Useful heat absorbed by the block (50%) is $\Delta Q = 0.75 \times 10^6\,\mathrm{J}$. Given $m = 8.0\,\mathrm{kg}$ and $s_{\mathrm{Al}} = 0.91\,\mathrm{J\,g^{-1}\,K^{-1}} = 910\,\mathrm{J\,kg^{-1}\,K^{-1}}$:

$$\Delta T = \frac{\Delta Q}{m s_{\mathrm{Al}}} = \frac{0.75 \times 10^6}{8.0 \times 910} = \frac{750,000}{7280} \approx 103.0\,^{\circ}\mathrm{C}$$

Practice Question 8 — Calorimetry with Phase Change & Heat Loss Error Analysis

A copper block of mass $2.5\,\mathrm{kg}$ is heated in a furnace to $500\,^{\circ}\mathrm{C}$ and then placed on a large ice block at $0\,^{\circ}\mathrm{C}$ ($s_{\mathrm{Cu}} = 0.39\,\mathrm{J\,g^{-1}\,K^{-1}}$, $L_{f,\mathrm{water}} = 335\,\mathrm{J\,g^{-1}}$). What is the maximum mass of ice that can melt? Additionally, in a standard calorimeter experiment where a hot metal block is dropped into water to determine the specific heat s_m of the metal, if heat losses to the surroundings are *not* negligible, is the experimentally computed value of s_m greater or smaller than the actual value?

(A) $1.455\,\mathrm{kg}$; smaller than the actual value

(B) $1.455\,\mathrm{kg}$; greater than the actual value

(C) $2.910\,\mathrm{kg}$; smaller than the actual value

(D) $0.728\,\mathrm{kg}$; greater than the actual value

Correct Answer: (A)

Worked Solution:

1. Maximum mass of ice melted:

Heat released by the $2.5\,\mathrm{kg}$ ($2500\,\mathrm{g}$) copper block cooling from $500\,^{\circ}\mathrm{C}$ to $0\,^{\circ}\mathrm{C}$ is:

$$Q = m_{\mathrm{Cu}} s_{\mathrm{Cu}} \Delta T = 2500\,\mathrm{g} \times 0.39\,\mathrm{J\,g^{-1}\,K^{-1}} \times 500\,\mathrm{K} = 487,500\,\mathrm{J}$$

Mass of ice melted:

$$m_{\mathrm{ice}} = \frac{Q}{L_f} = \frac{487,500\,\mathrm{J}}{335\,\mathrm{J\,g^{-1}}} \approx 1455.2\,\mathrm{g} = 1.455\,\mathrm{kg}$$

2. Effect of heat loss to surroundings:

If some heat is lost to the surroundings, the observed final equilibrium temperature T_f of the calorimeter water is slightly lower than it would be in an ideal isolated system. Because the computed heat gained by the calorimeter + water is smaller, the calculated specific heat of the metal is **smaller than the actual value**.

BOSONANCE

Practice Question 9 — Physiological Cooling by Evaporation of Sweat

A child of mass $30\,\text{kg}$ running a fever of $101\,^\circ\text{F}$ is given an antipyretic medicine that increases the rate of evaporation of sweat from his body. If the fever is brought down to $98\,^\circ\text{F}$ in $20\,\text{minutes}$, what is the average rate of extra evaporation caused by the drug? Assume evaporation is the only mechanism of heat loss, the specific heat capacity of the human body equals that of water ($1\,\text{cal}\,\text{g}^{-1}\,^\circ\text{C}^{-1}$), and the latent heat of evaporation of water at body temperature is $580\,\text{cal}\,\text{g}^{-1}$.

(A) $4.31\,\text{g}\,\text{min}^{-1}$

(B) $8.62\,\text{g}\,\text{min}^{-1}$

(C) $2.15\,\text{g}\,\text{min}^{-1}$

(D) $7.76\,\text{g}\,\text{min}^{-1}$

Correct Answer: (A)

Worked Solution:

First convert the temperature drop from Fahrenheit to Celsius:

$$\Delta t_F = 101\,^\circ\text{F} - 98\,^\circ\text{F} = 3\,^\circ\text{F} \implies \Delta t_C = \frac{5}{9}\Delta t_F = \frac{5}{9}\times 3 = \frac{5}{3}\,^\circ\text{C}$$

Total heat that must be removed from the child ($m = 30\,\text{kg} = 30,000\,\text{g}$):

$$Q = ms\Delta t_C = 30,000\,\text{g}\times 1\,\text{cal}\,\text{g}^{-1}\,^\circ\text{C}^{-1}\times\frac{5}{3}\,^\circ\text{C} = 50,000\,\text{cal}$$

If mass m_{sweat} evaporates with latent heat $L_v = 580\,\text{cal}\,\text{g}^{-1}$:

$$m_{\text{sweat}} = \frac{Q}{L_v} = \frac{50,000\,\text{cal}}{580\,\text{cal}\,\text{g}^{-1}} \approx 86.21\,\text{g}$$

Over $t = 20\,\text{min}$, the average rate of extra evaporation is:

$$\frac{m_{\text{sweat}}}{t} = \frac{86.21\,\text{g}}{20\,\text{min}} \approx 4.31\,\text{g}\,\text{min}^{-1}$$

BOSONANCE

Practice Question 10 — Steady-State Conduction in a Cubical Icebox and a Brass Boiler

Part I: A cubical thermacole icebox of side $30\,\text{cm}$ has a wall thickness of $5.0\,\text{cm}$ ($K = 0.01\,\text{W m}^{-1}\,\text{K}^{-1}$). If $4.0\,\text{kg}$ of ice at $0\,^\circ\text{C}$ is placed inside when the outside temperature is $45\,^\circ\text{C}$, how much ice remains unmelted after $6\,\text{hours}$? ($L_f = 335 \times 10^3\,\text{J kg}^{-1}$).

Part II: A brass boiler has a base area of $0.15\,\text{m}^2$ and thickness $1.0\,\text{cm}$ ($K_{\text{brass}} = 109\,\text{W m}^{-1}\,\text{K}^{-1}$). It boils water at $100\,^\circ\text{C}$ at the rate of $6.0\,\text{kg/min}$ ($L_v = 2256 \times 10^3\,\text{J kg}^{-1}$). What is the temperature of the part of the flame in contact with the boiler?

(A) Ice remaining $= 3.69\,\text{kg}$; Flame temperature $= 238\,^\circ\text{C}$

(B) Ice remaining $= 0.31\,\text{kg}$; Flame temperature $= 238\,^\circ\text{C}$

(C) Ice remaining $= 3.69\,\text{kg}$; Flame temperature $= 138\,^\circ\text{C}$

(D) Ice remaining $= 3.13\,\text{kg}$; Flame temperature $= 338\,^\circ\text{C}$

Correct Answer: (A)

Worked Solution:

Part I (Cubical Icebox): Total surface area of the 6 faces of the cube of side $s = 0.30\,\text{m}$ is $A = 6s^2 = 6(0.30)^2 = 0.54\,\text{m}^2$. Thickness $L = 0.05\,\text{m}$, $\Delta T = 45\,^\circ\text{C}$, time $t = 6\,\text{h} = 21,600\,\text{s}$. Total heat entering the box:

$$Q = \frac{KA\Delta T}{L}t = \frac{0.01 \times 0.54 \times 45}{0.05} \times 21,600 = 4.86\,\text{W} \times 21,600\,\text{s} = 104,976\,\text{J}$$

Mass of ice melted: $m_{\text{melted}} = \dfrac{Q}{L_f} = \dfrac{104,976}{335 \times 10^3} \approx 0.313\,\text{kg}$.

Mass of ice remaining $= 4.0 - 0.313 = 3.687\,\text{kg} \approx 3.69\,\text{kg}$.

Part II (Brass Boiler): Rate of heat required to boil $6.0\,\text{kg/min} = 0.10\,\text{kg/s}$ of water:

$$H = \frac{\text{d}m}{\text{d}t}L_v = 0.10\,\text{kg s}^{-1} \times 2256 \times 10^3\,\text{J kg}^{-1} = 2.256 \times 10^5\,\text{W}$$

Using $H = \dfrac{KA(T_1 - 100)}{L}$ with $A = 0.15\,\text{m}^2$ and $L = 0.01\,\text{m}$:

$$T_1 - 100 = \frac{HL}{KA} = \frac{2.256 \times 10^5 \times 0.01}{109 \times 0.15} = \frac{2256}{16.35} \approx 138\,^\circ\text{C} \implies T_1 = 238\,^\circ\text{C}$$

BOSONANCE

Practice Question 11 — Conceptual Reasoning on Thermal Radiation & Heat Transfer

Consider the following four physical statements:
1. A body with a large reflectivity is a poor emitter of thermal radiation.
2. A brass tumbler feels much colder than a wooden tray on a chilly day because brass has a much higher specific heat capacity than wood.
3. An optical pyrometer calibrated for ideal blackbody radiation gives too low a value for the temperature of a red-hot iron piece in the open, but gives a correct value when the same piece is inside a furnace.
4. Heating systems based on circulation of steam at $100\ ^\circ\mathrm{C}$ are more efficient in warming a building than those based on circulation of hot water at $100\ ^\circ\mathrm{C}$.
Which of the above statements are **correct**?

(A) 1, 3, and 4 only

(B) 1, 2, and 3 only

(C) 2 and 4 only

(D) 1, 2, 3, and 4

Correct Answer: (A)

Worked Solution:

- **Statement 1 is correct:** A body with large reflectivity is a poor absorber of radiation, and good absorbers are good emitters while poor absorbers are poor emitters.
- **Statement 2 is incorrect:** A brass tumbler feels much colder than a wooden tray on a chilly day because brass has a much higher *thermal conductivity* (not specific heat capacity) than wood, so it conducts heat away from the hand rapidly.
- **Statement 3 is correct:** In the open, a red-hot iron piece radiates at a lower rate $H = e\sigma A(T^4 - T_0^4)$ with $e < 1$, whereas inside a uniform-temperature furnace it reflects as well as emits radiation and behaves like a full blackbody radiator at temperature T.
- **Statement 4 is correct:** Steam at $100\ ^\circ\mathrm{C}$ carries an extra $22.6 \times 10^5\ \mathrm{J\,kg^{-1}}$ of latent heat of vaporisation compared to liquid water at $100\ ^\circ\mathrm{C}$.

BOSONANCE

Practice Question 12 — Newton's Law of Cooling Across Two Intervals

A body cools from $80\,^\circ\text{C}$ to $50\,^\circ\text{C}$ in $5\,\text{minutes}$. Calculate the time it takes to cool from $60\,^\circ\text{C}$ to $30\,^\circ\text{C}$ when the temperature of the surroundings is $20\,^\circ\text{C}$ (using the standard average-temperature method).

(A) $9\,\text{minutes}$

(B) $7.5\,\text{minutes}$

(C) $10\,\text{minutes}$

(D) $12\,\text{minutes}$

Correct Answer: (A)

Worked Solution:

First interval ($80\,^\circ\text{C} \to 50\,^\circ\text{C}$ in $t_1 = 5\,\text{min}$):

Average temperature $T_{\text{avg},1} = \dfrac{80+50}{2} = 65\,^\circ\text{C}$. Excess over surroundings ($T_s = 20\,^\circ\text{C}$) is $65 - 20 = 45\,^\circ\text{C}$:

$$\frac{80-50}{5\,\text{min}} = K(45\,^\circ\text{C}) \implies 6 = 45K \quad \text{--- (1)}$$

Second interval ($60\,^\circ\text{C} \to 30\,^\circ\text{C}$ in time t_2):

Average temperature $T_{\text{avg},2} = \dfrac{60+30}{2} = 45\,^\circ\text{C}$. Excess over surroundings is $45 - 20 = 25\,^\circ\text{C}$:

$$\frac{60-30}{t_2} = K(25\,^\circ\text{C}) \implies \frac{30}{t_2} = 25K \quad \text{--- (2)}$$

Dividing equation (1) by equation (2):

$$\frac{6}{30/t_2} = \frac{45}{25} \implies \frac{t_2}{5} = \frac{9}{5} \implies t_2 = 9\,\text{minutes}$$

8. Common Pitfalls & Quick Revision Checklist

- **Pitfall 1 — Temperature vs. Temperature Interval:** A specific temperature reading converts via $T = t_C + 273.15$ and $t_F = \frac{9}{5}t_C + 32$, whereas a *temperature difference* converts via $\Delta T = \Delta t_C$ and $\Delta t_F = 1.8\,\Delta t_C$. Never add 273.15 or 32 to a temperature difference ΔT!
- **Pitfall 2 — Expansion of Cavities and Holes:** When a metal plate with a circular hole is heated, the hole *increases* in diameter by $\Delta D = D\alpha_l \Delta T$, exactly as if it were filled with the same metal.
- **Pitfall 3 — Incomplete Melting or Vaporisation in Calorimetry:** Always check whether the available heat from the warmer substance is sufficient to melt all the ice before assuming a final temperature above $0\,^\circ\text{C}$. If available heat $Q_{\text{avail}} < m_{\text{ice}} L_f$, the final mixture remains at $0\,^\circ\text{C}$ with a fraction of ice unmelted.
- **Pitfall 4 — Conduction vs. Convection in Fluids:** A hot metallic bar placed under a running tap loses heat at its surface by *conduction* between the bar surface and the touching water layer, and then the moving water carries that heat away.
- **Pitfall 5 — Absolute Temperature in Radiation Laws:** Always convert Celsius temperatures to Kelvin ($T = t_C + 273.15$) before applying Wien's law ($\lambda_m T = b$) or the Stefan–Boltzmann law ($H = e\sigma A(T^4 - T_s^4)$), and convert wavelengths from ångströms ($1\,\text{Å} = 10^{-10}\,\text{m}$) or micrometres ($1\,\mu\text{m} = 10^{-6}\,\text{m}$) to metres.

BOSONANCE

9. Concise Final Formula Recap

Topic	Key Equation / Result	SI Units & Constants
Scale Conversions	$\frac{t_F - 32}{180} = \frac{t_C}{100}, \quad T = t_C + 273.15$	Triple point of water: $T_{\text{tr}} = 273.16\,\text{K}\ (0.01\,^\circ\text{C})$
Ideal Gas Law	$PV = \mu RT, \quad \alpha_v = \frac{1}{T}$ (at constant P)	$R = 8.31\,\text{J mol}^{-1}\,\text{K}^{-1}$
Thermal Expansion	$\frac{\Delta l}{l} = \alpha_l \Delta T, \quad \frac{\Delta A}{A} = 2\alpha_l \Delta T,$ $\frac{\Delta V}{V} = \alpha_v \Delta T = 3\alpha_l \Delta T$	α_l, α_v in K^{-1}; Thermal stress $= Y\alpha_l \Delta T$
Specific & Latent Heat	$\Delta Q = ms\Delta T = \mu C \Delta T, \quad Q_{\text{phase}} = mL$	Water: $s = 4186\,\text{J kg}^{-1}\,\text{K}^{-1}$, $L_f = 3.33 \times 10^5\,\text{J kg}^{-1}$, $L_v = 22.6 \times 10^5\,\text{J kg}^{-1}$
Heat Conduction	$H = KA\frac{T_C - T_D}{L}$, Series ($L_1 = L_2, A_1 = A_2$): $K' = \frac{2K_1K_2}{K_1 + K_2}$	H in W (J s^{-1}), K in $\text{W m}^{-1}\,\text{K}^{-1}$
Thermal Radiation	$\lambda_m T = b, \quad H_{\text{net}} = e\sigma A\left(T^4 - T_s^4\right)$	$b = 2.9 \times 10^{-3}\,\text{m K}$, $\sigma = 5.67 \times 10^{-8}\,\text{W m}^{-2}\,\text{K}^{-4}$
Newton's Law of Cooling	$-\frac{\text{d}Q}{\text{d}t} = k(T_2 - T_1),$ $\frac{T_i - T_f}{t} = K\left(\frac{T_i + T_f}{2} - T_1\right)$	Valid for small excess temperatures $(T_2 - T_1)$

BOSONANCE